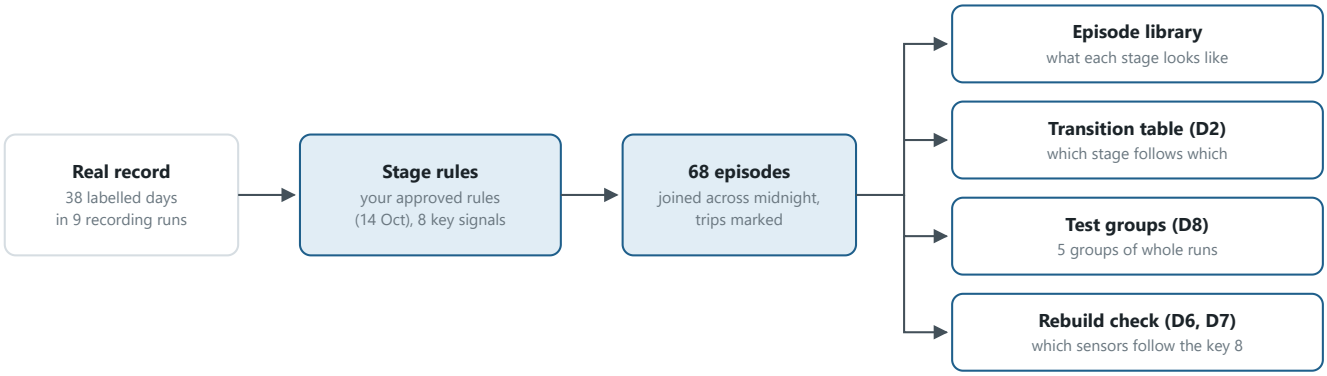
SOFC synthetic data · Step 1 results · 14 October 2026

The learning material is ready, and the data changes three parts of the plan

The answer in one sentence. The 38 labelled days are now cut into 68 stage episodes, and the material each later step learns from is built from them. The data shows that rated power never happened, part load is short and usually ended by a safety trip, and only 16 of 28 sensor groups can be rebuilt from the 8 key signals.

Why it matters. The plan assumed six stages with enough real examples of each. That holds for start-up, hot standby and shutdown, but not for rated power or a steady part load. Those need a different approach before any fake data is made.

What was built



The stage rules are the ones you approved on 14 October (operational-modes.md). Nothing here leaves ERC: these files are the inputs for Step 2.

What the data showed, and what it changes

Finding	What it changes
Rated power never happens	Highest load was 18.8 A per string (63% of the rated 30 A). There is nothing to learn it from. Version 1 either leaves it out or builds it from design values, labelled "not learned from data". **Your choice.**
Part load is short and trip-ended	All 10 load periods last 8 to 44 minutes, and 8 of them end in a safety event (emergency stop, purge or power cut) within 15 minutes. There is no long, steady part load to imitate. The fake part load must be built mainly from the ramp rules (up to 0.9 A/min per string) and marked as "low variety".
Enough start-ups, hot standbys and shutdowns	12 start-ups, 16 hot standbys and 7 shutdowns are usable. That is enough for the curve families of Step 2. The one-or-two-example fallback (D3) is not needed for these.
2 of 7 usable shutdowns start with a trip	Their cooling curves are kept as examples of an emergency shutdown, separate from planned shutdowns.
Two stage changes not in the design	A start-up sometimes stops and the stack cools again (twice back to cold, twice into a shutdown). The pattern maker should allow this "aborted start-up".
121 of 175 kept sensors follow the 8 key signals	Those can be rebuilt by prediction. The rest (gas supply, string voltages, reformer inlets, flow set points) need their own small generators or stage rules.
The 36 stack temperatures move as one	One shared shape explains 99.7% of their movement. Three shapes (the plan's minimum) are more than enough.

Sources: Synthetic data generation/library/ (episodes.csv, transitions.csv, rebuild_r2_per_group.csv, stack_shapes.json), built by build_library.py from data_analysis/operating_states.py.

If you remember one thing

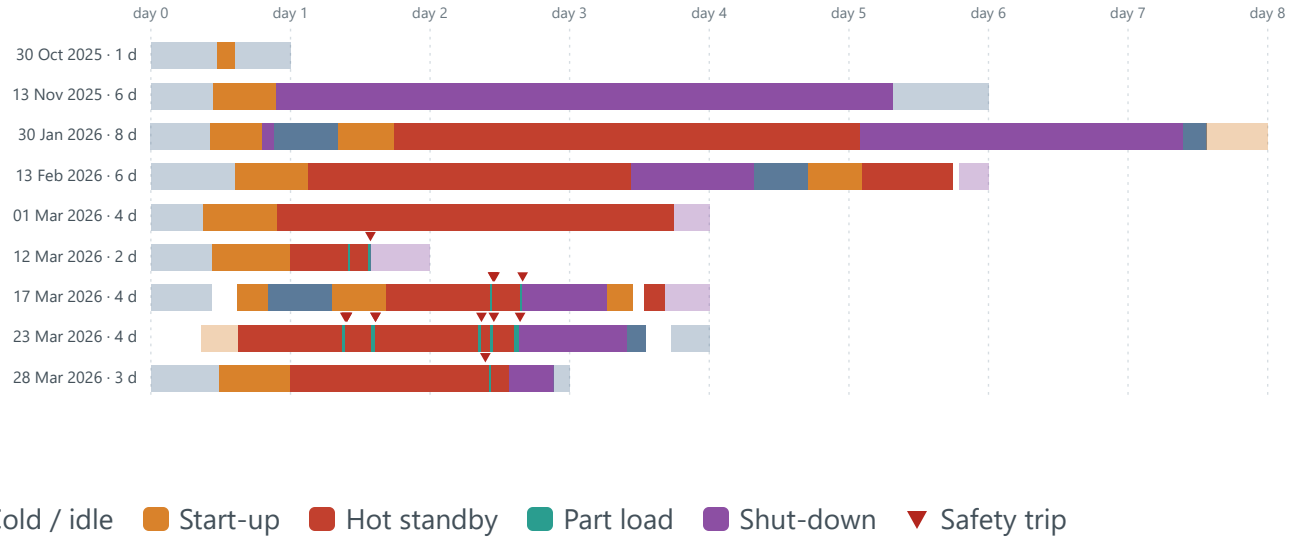
The plan works for start-up, hot standby and shutdown. Rated power and steady part load were never really recorded, so they need a decision, not a model.

The episode library

In one sentence. Every stage episode in the 38 days is now a separate record with its key-signal trace, so Step 2 can study, for example, all 12 start-ups side by side.

Why it matters. Fake start-ups and shutdowns are drawn from a family fitted to these real ones. A wrongly cut episode would teach the wrong shape.

The nine recording runs, coloured by stage



Cold / idle · Start-up · Hot standby · Part load · Shut-down · Safety trip

Faded = cut by the edge of a run, not used

Each row is a run of back-to-back days. Hover a block for its times. Part load shows only as thin slivers in March, mostly with a trip right after. 22 March has no 10-second file, so the 23 March run starts at 08:40.

How many examples each stage has

Stage	Episodes	Fully inside a run	Usable for learning
Cold / idle	17	5	**5**
Start-up	14	12	**12**
Hot standby	16	16	**16**
Part load	10	10	**10**
Rated power	0	0	**0**
Shut-down	11	7	**7**

An episode that runs into the start or end of a recording run is cut off, so its real length is unknown. Those are kept for reference but not learned from. Most cold periods are like this, which does not matter: a cold, idle stack is flat.

How safety trips are handled

Fifty-seven trip and cut events are listed in the record. Fourteen of them are real safety events (emergency stop, emergency purge, power cut at a trip value). Each episode is used only on its longest stretch that stays at least 10 minutes away from such an event (decision D19). A shutdown that begins with a trip is kept and labelled **trip-initiated**, because it is a real example of an emergency shutdown.

> **Where the files are.** `Synthetic data generation/library/episodes.csv` (one row per episode) and `traces_1min.parquet` (the 8 key signals per minute, with a "usable" flag). Rebuild them with `build_library.py`.

If you remember one thing

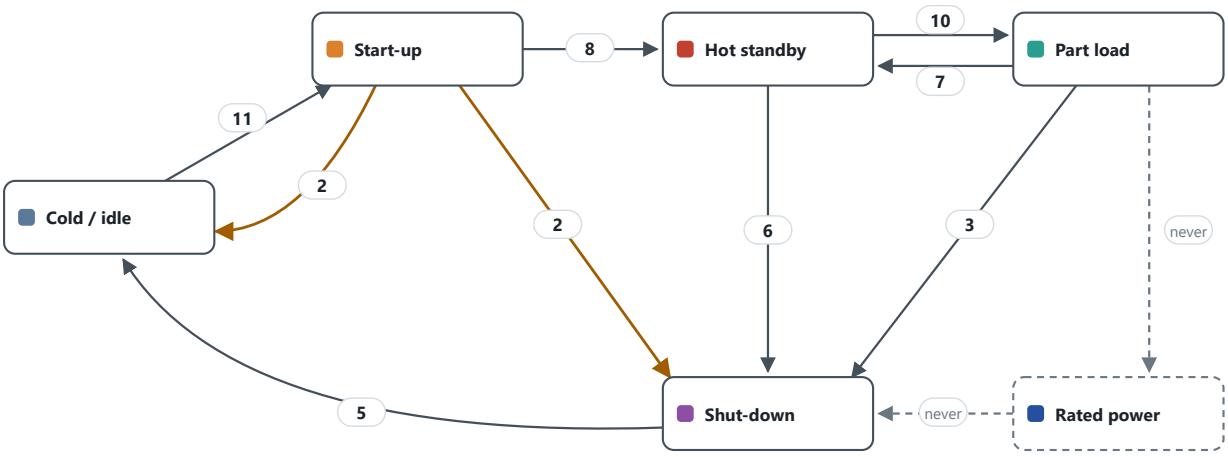
Start-up, hot standby and shutdown each have 7 to 16 clean examples. Part load has 10, but none longer than 44 minutes.

Which stage follows which

In one sentence. Counted from the data: the plant follows the design flow, except that a start-up is sometimes stopped before the stack is hot, and rated power never occurs.

Why it matters. The pattern maker only builds sequences this table allows. A missing arrow means the fake data can never show that change. An extra arrow means it may show something the plant never does.

Stage changes in the 38 days



Seen, and in the design — Seen, not in the design - - - In the design, never seen

Numbers are how many times each change happened. The design flow is from operational-modes.md §2. Changes across a logger gap of more than an hour are not counted.

Proposed rule for the pattern maker

- **Allow** every change seen in the data, including the two kinds of aborted start-up (stack cools back to cold, or a shutdown follows).
- **Allow from design only if you decide to include rated power:** part load ↔ rated power, and rated power → hot standby or shutdown, all marked "not learned from data".
- **Do not allow** anything else, for example hot standby straight back to start-up. Under your 14 October rule, a stack that has been in hot standby re-heats within hot standby, not as a new start-up.

▶ All counted changes

> **If you remember one thing**
>
> The data agrees with the design flow, adds one real behaviour (the aborted start-up), and never reaches rated power.

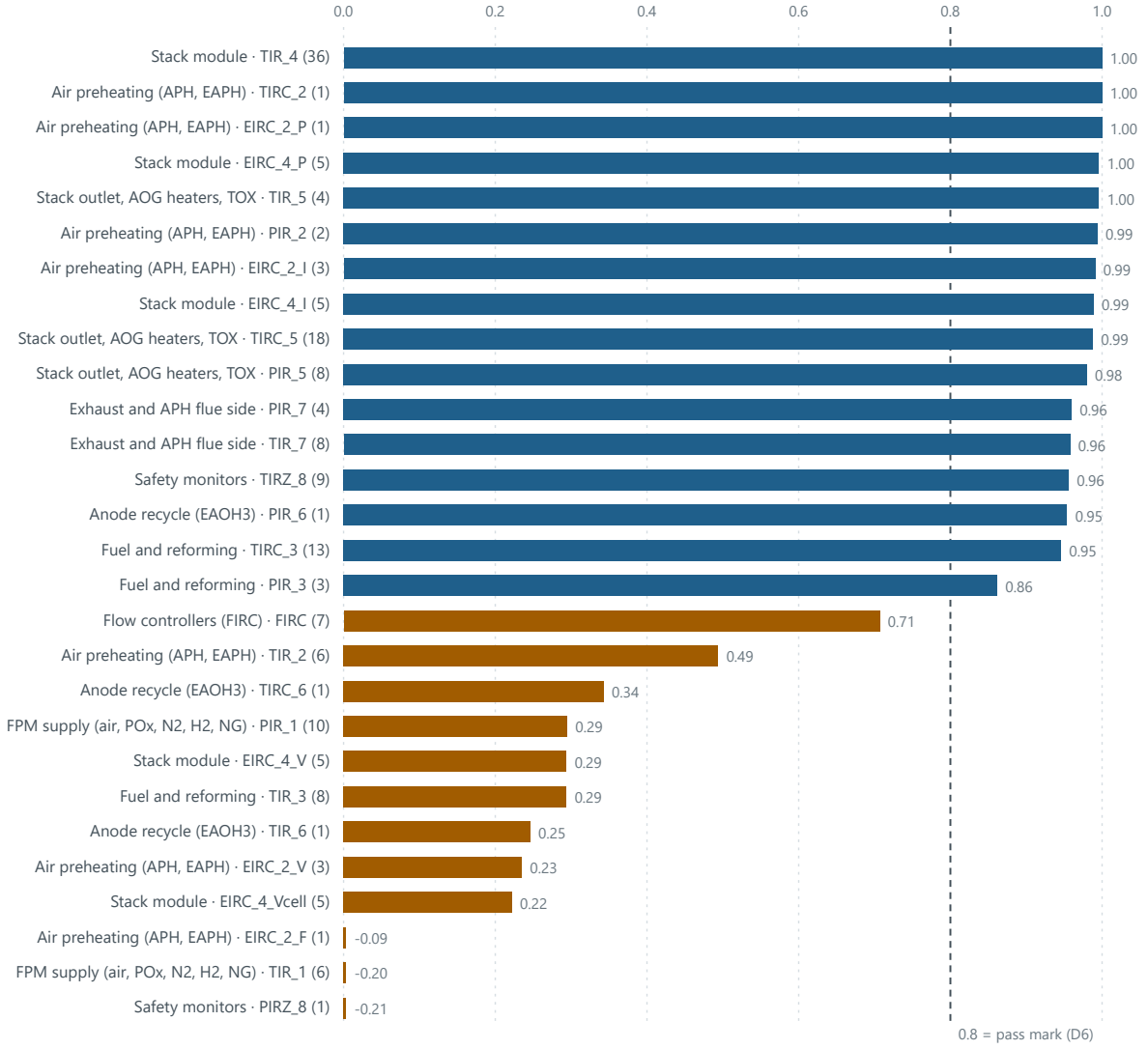
Output 4 of 4 · decisions D6 and D7

Which sensors can be rebuilt from the key eight

In one sentence. 121 of the 175 kept sensors (16 of 28 groups) can be predicted well from the 8 key signals. The other 54 need their own small generators.

Why it matters. The plan makes 8 signals first and predicts the rest. That only works for sensors that really follow those 8. This check says which ones do, measured on days the prediction never saw.

How well the 8 key signals predict each sensor group



Rebuild from key signals Needs its own generator or stage rule

R^2 is the share of a sensor's movement the prediction gets right: 1 is perfect, 0 is no better than its average (negative values are drawn at 0). Each bar is the median over the group's sensors (count in brackets). Scored on held-out test groups, never on the days used to fit.

What falls short, and why

Group	Likely reason	Proposed route
Gas supply pressures and temperatures (PIR 1, TIR 1)	Set by the supply side and the room, not by the stack	Slow trend plus noise, drawn per stage
String voltages, minimum cell voltage (EIRC 4)	Only valid while the load is on; follow a voltage–current curve	Simple physics: open-circuit voltage minus resistance × current, fitted per string
Flow controllers (FIRC)	Step-like set points chosen by the operator	Stage rules (which flow at which stage), plus measured noise
Reformer inlets, APH inlets, recycle (TIR 3, TIR 2, TIR 6, TIRC 6)	Depend on gas flows, not only on stack heat	Predict from the flow set points once those exist
EAPH mains voltage and frequency, PIRZ 8	Grid and room conditions	Steady value plus noise

About 290 of the 466 logged columns were already set aside in preprocessing as set points, status flags or constants. They follow stage rules, not prediction.

The 36 stack temperatures

34 working thermocouples (TIR_4_6_25 reads room temperature and is left out). One shared shape explains 99.7% of their movement and two explain 99.99%. The plan's minimum of three shapes (D7) is more than enough: one for the level, one for the hot–cold spread, one spare.

> **A caution.** The stack group scores almost perfectly partly because three key signals are themselves made from it (median, lowest, highest). The same holds for string power, which is voltage × current. The scores for other groups are the honest test.

If you remember one thing

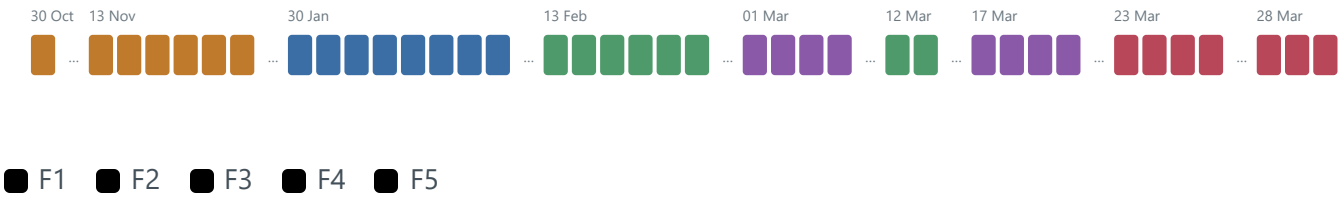
Two thirds of the sensors can be predicted from the key eight. The rest are supply-side, voltage or set-point signals that need simple rules of their own.

Test groups, decisions, and the next step

In one sentence. The 38 days are split into 5 test groups of whole recording runs, fixed now so no later model sees its test days in advance. Step 2 can start once you decide on rated power.

Why it matters. Days inside one run are near-copies of each other (the stack carries its heat across midnight). Splitting by run, not by day, stops a test from passing only because it saw yesterday.

The 38 days by test group



Each block is a day, grouped by run; "…" marks a break in recording. Load days exist only in March, so part load can be tested in three of the five groups (F3, F4, F5).

What each test group holds

Group	Days	Runs (first day)	Cold	Start-up	Hot st.	Part load	Shut-down
F1	8	2026-01-30	2	2	1	0	2
F2	7	2025-10-30, 2025-11-13	0	2	0	0	1
F3	8	2026-02-13, 2026-03-12	1	3	4	2	1
F4	8	2026-03-01, 2026-03-17	1	4	4	2	1
F5	7	2026-03-23, 2026-03-28	1	1	7	6	2

Counts are usable episodes. F2 holds the two oldest runs (October and November), so it also shows whether the plant changed over time (D18).

Where the decisions stand

#	Decision	Status
D1	Stage sensors and thresholds	**Done** your rules of 14 Oct; the second labelling method and your check by eye of each start-up and shutdown are still open
D2	Which stage may follow which	**Proposed** from the data, plus the aborted start-up
D3	Stages with one or two examples	**Your call** only rated power is affected: leave out, or design values
D6	Rebuild route per group	**Proposed** 16 groups from key signals, the rest by own rules
D7	Number of temperature shapes	**Done** 3 (one already explains 99.7%)
D8	Held-out test days	**Done** 5 groups of whole runs (above)
D19	Faults and alarms	**Done** 14 safety events, kept 10 min clear; trip-initiated shutdowns kept as emergency examples

Next: Step 2, making each stage

1. **Start-up and shutdown curve families.** Fit heating and cooling shapes to the 12 start-ups and 7 shutdowns (planned and trip-initiated separately), then draw new curves kept away from every real one.
2. **Hot standby, two levels.** A slow trend plus a fitted wobble for the ~510 °C hold and the 575–610 °C standby.
3. **Part load from rules.** Ramps of up to 0.9 A/min per string, plateaus drawn from the 10 real ones, marked "low variety".
4. **First check.** Compare a fake start-up with the real ones in its held-out group.

If you remember one thing

The groundwork is done and the test days are fixed. One decision is needed from you, rated power, before Step 2 starts.